GAZI UNIVERSITY

C++

PRACTICAL FILE

Variables, Operators, Type Casting & Programs

INTRODUCTION

NAME :	Nisha Fatima
DEPARTMENT :	Computer Science
SECTION :	D
UNIVERSITY :	Gazi University

I am **Nisha Fatima**, a student of **Computer Science** in **Section D** at Gazi University. This file is a collection of my C++ programming work. It covers how to declare variables, take input with **cin**, show output with **cout**, solve expressions using operator precedence, use type casting, and build small programs such as a calculator, an average finder and a Math Tutor.

"Practice makes a programmer perfect."

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Task 1 : Variables, Input and Output

About this program

Three variables are defined: **age** (int), **pay** (double) and **section** (char). The program reads all three with **cin** and prints each on a new line with **cout**.

Program (Code)

```cpp
#include <iostream>
using namespace std;
int main()
{
    int age;
    double pay;
    char section;
    cin >> age >> pay >> section;
    cout << age << endl;
    cout << pay << endl;
    cout << section << endl;
    return 0;
}
```

Output

```
Example input:   18 25000.50 A

Output:
18
25000.5
A
```

Task 2 : Value of Each Expression

About this task

C++ follows operator precedence: brackets first, then *** / %** (left to right), then **+ -**. With int values, division throws away the decimal part and **%** gives the remainder.

No.	Expression	Working	Value
1	`28 / 4 - 2`	`7 - 2`	**5**
2	`6 + 12 * 2 - 8`	`6 + 24 - 8`	**22**
3	`4 + 8 * 2`	`4 + 16`	**20**
4	`6 + 17 % 3 - 2`	`6 + 2 - 2`	**6**
5	`2 + 22 * (9 - 7)`	`2 + 22 * 2`	**46**
6	`(8 + 7) * 2`	`15 * 2`	**30**
7	`(16 + 7) % 2 - 1`	`23 % 2 - 1 = 1 - 1`	**0**
8	`12 / (10 - 6)`	`12 / 4`	**3**
9	`(19 - 3) * (2 + 2) / 4`	`16 * 4 / 4`	**16**

Correction

Expression 4: 17 % 3 = 2, so 6 + 2 - 2 = **6** (not 7).

Task 3 : Automatic Data Type Conversion

Answer (c)

units is promoted to a float, **mass** remains a float, and the result of **mass * units** is a float. When an int and a float are used together, C++ automatically promotes the int to the bigger type.

Task 4 : What value will be stored in a ?

Correct Answer : B. 8

1. b = 2 (integer)
2. c = 4.2 (float)
3. b * c = 2 * 4.2 = 8.4
4. Since **a** is an int, the decimal part is discarded when 8.4 is stored in it. Therefore **a = 8**.

Task 5 : Type Cast Expression

About this task

qty and salesReps are both integers, so **qty / salesReps** performs integer division and loses the decimal part. A type cast fixes this.

Correct Answer

```
unitsEach = static_cast<float>(qty) / salesReps;
```

Explanation

- qty and salesReps are both integers.
- Integer division discards the decimal part of the result.
- static_cast<float>(qty) converts qty into a float.
- Therefore the division produces a floating-point result instead of integer division.

Task 6 : Math Tutor

About this program

This program works like a small teacher. It makes two random 3-digit numbers using **rand() % 900 + 100**, shows them as an addition problem and waits for the student to press Enter. After that it prints the problem again with the correct answer. **srand(time(0))** makes the numbers different every time the program runs. (Example from the task: 247 + 129 = 376.)

Program (Code)

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

int main()
{
    int num1, num2, answer;
    srand(time(0));
    num1 = rand() % 900 + 100;
    num2 = rand() % 900 + 100;

    cout << "Math Tutor" << endl;
    cout << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "--------" << endl;
    cout << "Press Enter to check the answer...";
    cin.get();
    answer = num1 + num2;
    cout << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "--------" << endl;
    cout << answer << endl;
    return 0;
}
```

Output (screenshot)

```
Math Tutor
641
+ 665
--------
Press Enter to see the answer...
641
+ 665
--------
1306

[Program finished]
```

HOME TASK

Program 1 : Sum of Two Numbers

About this program

The user enters two numbers (stored as **double**, so decimals also work). The program adds them and displays the sum.

Program (Code)

```cpp
#include <iostream>
using namespace std;

int main()
{
    double number1, number2, sum;

    cout << "Enter a number: ";
    cin >> number1;

    cout << "Enter another number: ";
    cin >> number2;

    sum = number1 + number2;

    cout << "The sum of the two numbers is "
         << sum << endl;

    return 0;
}
```

Output (screenshot)

```
Enter a number: 12
Enter another number: 23
The sum of the two numbers is 35

[Program finished]
```

Program 2 : Division using static_cast

About this program

Two integers are read and divided. Because **number1** is cast to float with **static_cast<float>**, the answer keeps its decimal part (for example 7 / 2 gives 3.5 instead of 3).

Program (Code)

```cpp
#include <iostream>
using namespace std;

int main()
{
    int number1, number2;
    float quotient;

    cout << "Enter two numbers and I will divide\n";
    cout << "the first by the second for you.\n";

    cin >> number1 >> number2;

    quotient = static_cast<float>(number1) / number2;

    cout << quotient << endl;

    return 0;
}
```

Output (screenshot)

```
Enter two numbers and I will divide
the first by the second for you.
12 3
4

[Program finished]
```

Program 3 : Sum and Average of Five Numbers

About this program

Five integers (28, 32, 37, 24 and 33) are added to find the sum. The average is found by dividing the sum by **5.0** (a decimal number) so the result is not rounded down. Sum = 154 and Average = 30.8.

Program (Code)

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a = 28, b = 32, c = 37, d = 24, e = 33;
    int sum;
    float average;

    sum = a + b + c + d + e;

    average = sum / 5.0;

    cout << "Sum = " << sum << endl;
    cout << "Average = " << average << endl;

    return 0;
}
```

Output (screenshot)

```
Sum = 154
Average = 30.8

[Program finished]
```

Program 4 : Area of a Circle

About this program

The program reads the radius of a circle and finds its area with the formula **area = pi * r * r**. The value of pi (3.14) is stored in a **const float**, so it cannot be changed by mistake.

Program (Code)

```cpp
#include <iostream>
using namespace std;

int main()
{
    float r, area;
    const float pi = 3.14;

    cout << "Enter radius: ";
    cin >> r;

    area = pi * r * r;

    cout << "Area of circle = " << area;

    return 0;
}
```

Output

```
Example (radius = 5):
Enter radius: 5
Area of circle = 78.5
```

~ End of File ~

Prepared by Nisha Fatima | Computer Science, Section D | Gazi University